

Software Reverse Engineering Course

- Site: OnHexGroup.ir
- Youtube: @onhexgroup
- Telegram: onhex_ir
- Twitter: @onhexgroup
- Github: onhexgroup

Windows
(x86 / x64)

By: OnhexGroup

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How a Program Executes in Windows

Executable File Is Launched

- **By double-clicking**
- **From the command line**
- **By another program**
- **And other methods**

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How a Program Executes in Windows

Executable File Is Launched

**Windows Loader Examines the
File Header**

- **The Windows loader is responsible for loading the file into memory and preparing it for execution**
- **Executable files in Windows are called PE (Portable Executable) files**
- **PE files consist of multiple parts, including headers**

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**Windows Loader Examines the
File Header**

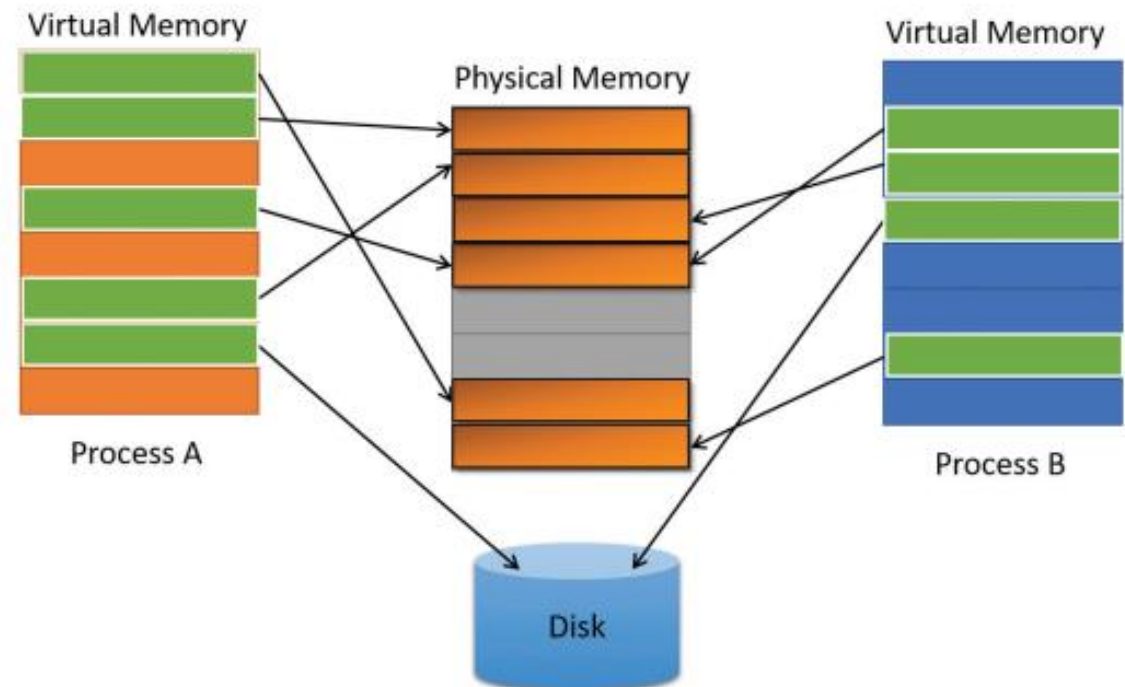
A Process Is Created

- **The loader creates a process and assigns an address space to it**
- **A process is like a bookshelf that holds all program components**

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Virtual Memory

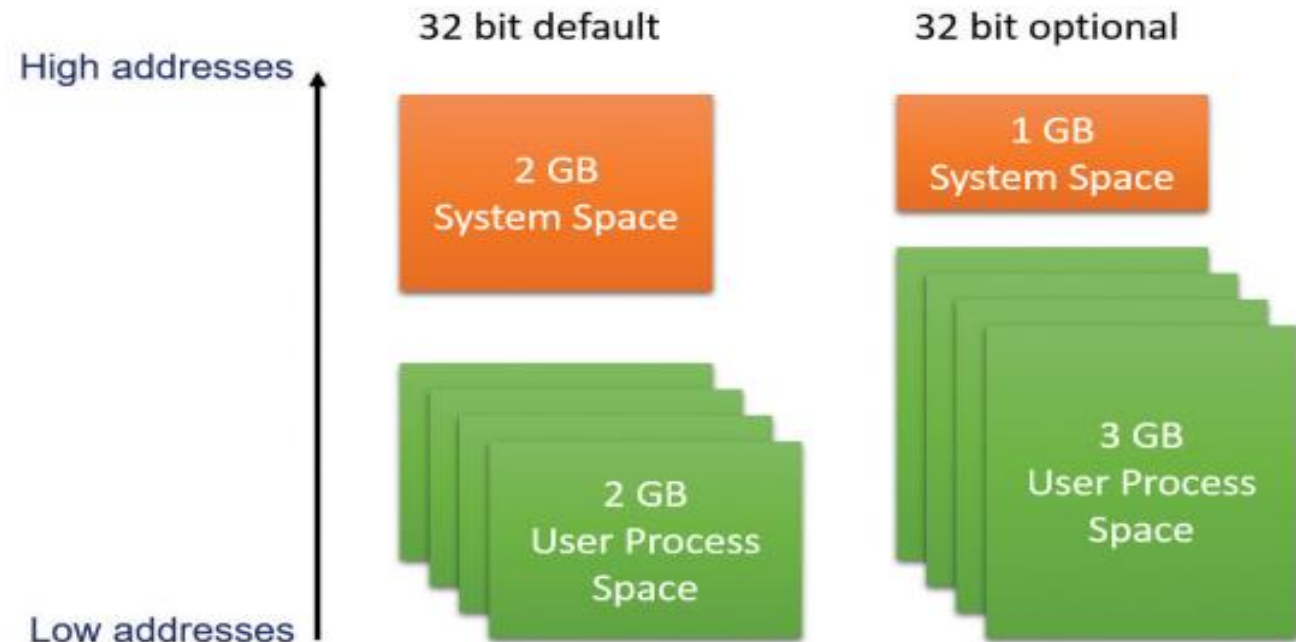
- **From the process perspective, physical memory does not exist; everything uses virtual memory**
- **The Memory Manager is responsible for: Reading and writing memory, Translating virtual addresses to physical addresses**



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Virtual Address Space

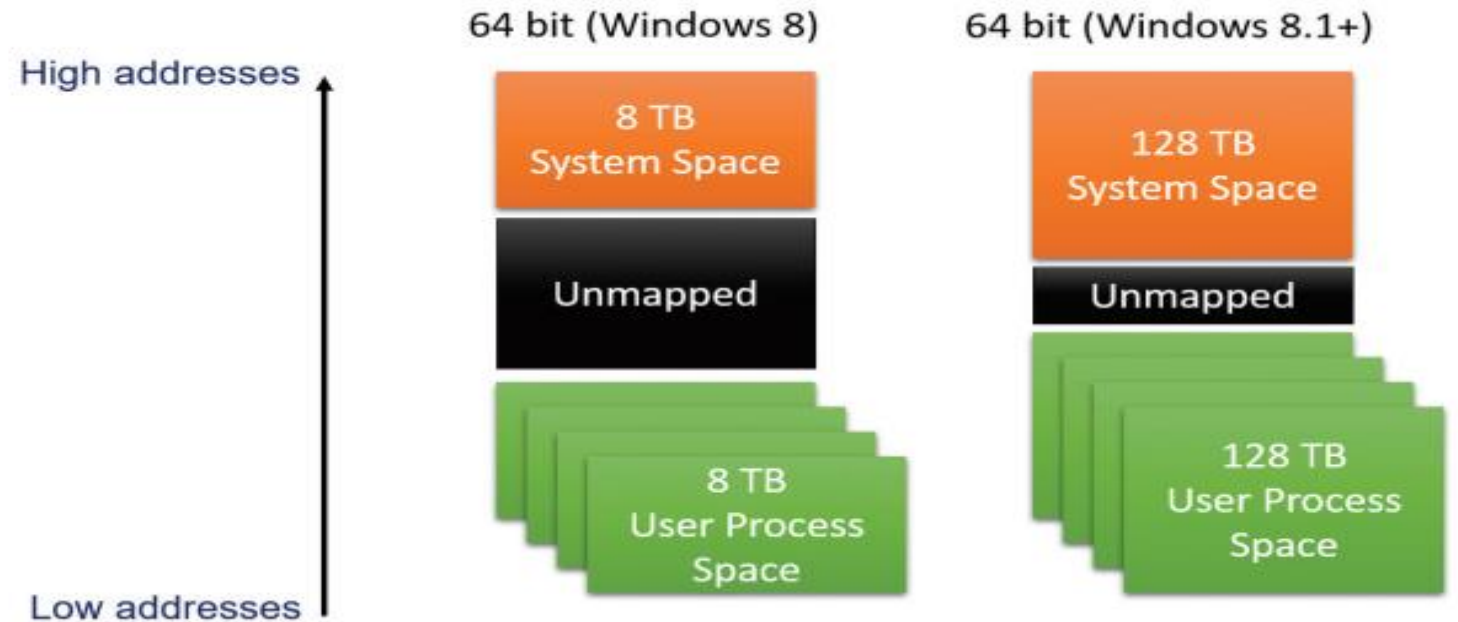
- **Each process has its own address space:**
 - **32-bit systems:** 2^{32} bytes (4 GB total)
 - **User space:**0x00000000 – 0x7FFFFFFF (2 GB)
 - **Kernel space:**0x80000000 – 0xFFFFFFFF (2 GB)



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Virtual Address Space

- **Each process has its own address space:**
 - **64-bit systems:** 2^{64} addresses (16 exabytes)
 - **User space:** `0x0000000000000000` – `0x00007FFFFFFFFFFFFF`
 - **Kernel space:** `0x0000800000000000` – `0xFFFFFFFFFFFFFFFF`



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How a Program Executes in Windows

Executable File Is Launched

**Windows Loader Examines the
File Header**

A Process Is Created

Loader Maps PE Sections

The loader maps each section to the appropriate location in memory

- **Each executable file contains multiple sections: Code, Data, Resources, ...**
- **CPU instructions are fetched from memory for execution**

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How a Program Executes in Windows

Executable File Is Launched

**Windows Loader Examines the
File Header**

A Process Is Created

Loader Maps PE Sections

Execution Transfers to the EP

- **Execution begins at the Entry Point (EP)**
- **The EP is defined in the PE header**

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Resources

- <https://learn.microsoft.com/en-us/windows-hardware/drivers/debugger/user-space-and-system-space>
- <https://learn.microsoft.com/en-us/windows-hardware/drivers/debugger/accessing-memory-by-virtual-address>
- <https://learn.microsoft.com/en-us/windows-hardware/drivers/gettingstarted/virtual-address-spaces>
- <https://learn.microsoft.com/en-us/windows/win32/memory/memory-limits-for-windows-releases#memory-and-address-space-limits>